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The hypothesis will be tested by accomplishing the following Specific Tasks: 1) To determine whether the reduction in apoptosis and proliferation in MIP-101 cells with simulated microgravity also occurs in other CRC cell lines and is associated with changes in the activation of signal transduction pathways that affect cell survival and proliferation, e.g., the MAP Kinase, STAT3, and phosphatidylinositol-3 kinase (PI3 Kinase) signal transduction pathways. 2) To determine whether the simulated microgravity of suspension culture affects the ability of CRC cells to survive oxidative stress in studies that will test our hypothesis that reduced gravity may alter metastatic potential as well as to establish the kinetics of CRC genetic reprogramming in response to reduced gravity. 3) To compare the pattern of gene expression in MIP-101, Clone A, SW620 and CX-1 cells and clones 16E and 19A of HT-29 induced by simulated microgravity to the profile of genes expressed by clones 16E or 19A at terminal differentiation. This will be used for selecting the appropriate CRC cell lines for a subsequent flight experiment based on determination of Euclidean distance for selected sets of signal transduction and oxidative stress genes. The objective of this project is to test the hypothesis that

Task Description:	three-dimensional (3-D) growth provided by simulated or actual microgravity facilitates the reprogramming of signal transduction pathways and the profiles of genes expressed by human colorectal carcinoma (CRC) cell lines as they differentiate toward either of the two major colonic cell lineages (goblet or absorptive cells). This differentiation is important because it may inhibit cancer growth and if applied to developing cancers may block the emergence of new cancers in people. Our postulate is that 3-D growth facilitates differentiation that otherwise occurs slowly in two-dimensional tissue culture either during spontaneous differentiation of certain human CRC lines or with various agents that have potential for chemoprevention of bowel cancer. These studies provide a critical link between the effects of microgravity and the gene expression programs and signal transduction pathways that are associated with spontaneous and chemical differentiation. Although reductions in apoptosis, proliferation and growth factor expression occur in actual microgravity, no specific analysis has sought to link the gene expression profile of cells in actual microgravity to the function of specific signal transduction pathways or to sensitivity to oxidative stress. Our studies aim to correct this deficit by focusing on the expression of genes involved in signal transduction and oxidative stress. We have also broadened our program to bring in the Augenlicht laboratory that is an international resource for the analysis of gene expression in human and murine CRC. Thus, this project has significant value to NASA and microgravity research because 1) it expands the expertise available to the microgravity community, 2) explores the possibility that gravity participates in metastasis and 3) seeks to link gene expression arrays to specific signal transduction pathways and oxidative stress protein function. Our long term goal is to identify those genes that regulate differentiation in human CRC cells so as to arrest cancer cell proliferation and promote cancer cell death.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work demonstrates that NO and ROS affect both epithelial cells and PBMC in a simulated microgravity environment to cause similar loss of cell function. We propose as a result that treatment with inhibitors of NO production that target iNOS may be critical to the outcome of situations where nitrosative stress is increased. These data are important to those studies that involve shock, sepsis, myocardial infarction and stroke on earth.
	1. Overview of Progress The applicant has been fortunate to have this grant entitled "Differentiation and Gene Expression of MIP-101 Carcinoma Cells" which is shared with Dr. L. Augenlicht, a molecular biologist at the Albert Einstein Cancer Center, managed by the Biological Systems Office of NASA/JSC. In addition, while the PI has worked to identify a putative mechanism for cell cycle arrest and apoptosis in the RWV, he used the RWV to analyze host-tumor interactions in a novel way that lead to 3 peer-reviewed publications (57, 59, 60) and two manuscripts recently published in Cancer Research and the International Journal of Cancer. We have now found that the poorly differentiated human colon carcinoma MIP-101 undergoes profound gene expression changes with shape change and fluid shear stress. These changes are noted below. Since we received half of our requested funding, we had to alter our specific aims accordingly. In addition, during the last two years of this grant we were requested to do two extra unfunded efforts: one a cell science experiment on ISS that was to be repeated on 13A.1 and then after the Columbia accident a series of mixing experiments with Dr. Zimmerberg and his team at NIH that comprises the CBOSS-FDI experiment. These unfunded requests as well as the decreased direct funding that barely covered the cost of a technician and supplies for the PI and co-PI reduced our ability to work on the original aims substantially. As a result, our focus shifted to identify the changes in gene expression that occur with shape change and exposure to fluid shear stress as well as the effects those changes have on the induction of apoptosis of both epithelial cells and immune cells. In addition, we have attempted to insure that our program begins to reorient itself to the Bioastronautics Critical Pathway for NASA and its future research programs by determining that the changes of intracellular nitrosative stress and apoptosis that we observe in our model cell system – the MIP-101 carcinoma – also occur in normal human PBMC. This indicates the value of our original model as an important bellweather for cellular changes in simulated microgravity. In addition, we have now identified that the actual cell death observed in rotational 3-D culture is due to activation of caspases that is a hallmark of apoptosis or programmed cell death and that is described below. In recent work we have developed reagents that specifically silence eNOS or iNOS and have shown that knock down of eNOS but not iNOS significantly decreases intracellular NO and increases apoptosis. 2. Importance of Rotational Anoikis as a Measure of the Metastatic Potential of Colon Cancer: ` With NASA support and the NASA-designed Rotating Wall Vessel (RWV), we have developed a method to assess the ability of human colorectal carcinoma cell lines to form metastases in the livers of athymic nude mice by determining whether the carcinoma cells survive 24 hr culture in the RWV. Normal, non-transformed cells do not survive under anchorage independent growth conditions but die from anoikis – a form of apoptosis caused by loss of integrin-mediated survival signals. In contrast, transformed or malignant cells survive in this anchorage independent, stationary 3-D culture environment (Static 3-D culture). Interestingly, laminar shear stress added to 3-D growth in suspension (i.e., Rotated 3-D culture) induces significantly more cell death in weakly than highly metastatic CRC. Viability is measured by flow cytometry of cells stained with Propidium Iodide (PI) and Annexin V-FITC with viable cells being Annexin V-/PI -. In contrast, weakly metastatic CRC are as or more viable under conditions that usually generate anoikis – stationary 3-D growth. Thus, viability in short term Rotated 3-D culture is associated with metastatic potential. This observation has led us to examine the mechanisms involved in rotational anoikis and then to determine whether they are important to metastasis. 3. Induction of Apoptosis in 3-D Rotated Cultures: Our studies initially used the poorly differentiated human CRC cell line MIP-101 that is weakly metastatic in the intrasplenic injection experimental metastasis assay. MIP-101 cells were cultured in 3-D with rotation (Rotated 3-D), in 3-D spheroid cultures over a nonadherent, PolyHEMA-coated surface (Static 3-D) or in standard 2-D monolayer culture on tissue culture plastic (Monolayer 2-D). Apoptosis was confirmed by flow cytometry with Annexin-V-FITC/PI staining and static cell fluorometry for DEVD degradation by activated caspase 3 (104). MIP-101 cells cultured in the RWV contain 56.3% Annexin V + cells as early as 24 hr after culture initiation whereas the monolayer and Static 3-D cultures displayed significantly lower levels of Annexin V+ cells (6.3% and 13.1%, respectively). Similarly, when MIP-101 cells were prelabeled with PhiPhiLux, a caspase 3 fluorogenic precursor substrate, a greater fraction of Caspase 3 active cells were observed in the Rotated 3-D cultures (29 + 8.2%) than in the Monolayer cultures (1 + 0.4%) or Static 3-D cultures (1.5 + 1.2%) (Mean + SEM, with $P<0.004$ for Rotated 3-D vs either of the other culture conditions). DNA damage also occurs because 47 ± 8% of MIP-101 cells in the Rotated 3-D cultures were positive by TUNEL assay (Trevigen) while the Monolayer 2-D and Static 3-D cultures were negative. When cell cultures were treated with the anti-oxidant N-acetylcysteine (NAC), the PARP inhibitor

Task Progress:

3-aminobenzamide (3-AB) or the caspase inhibitor Z-VAL-Ala(Ome)-Asp-Fluoromethylketone (Z-VAD-fmk), the caspase inhibitor blocked cell death while treatment with NAC or 3-AB increased death. Thus, MIP-101 cells, like most malignant cells, are resistant to static anoikis, but sensitive to death in rotational 3-D cultures. Further, short term Rotated 3-D cultures of MIP-101 cells undergo apoptosis that involves DNA damage, activation of caspase 3 and inhibition by treatment with Z-VAD-FMK. Rotational 3-D cultures undergo a subtype of apoptosis that we term rotational anoikis.

4. 3-D Rotational Culture Increases iNO and iROS: We have found that both iNO and iROS are increased within CRC as they transition from 2-D to 3-D stationary to Rotated 3-D culture and that iNO can be decreased by a general NOS inhibitor. Since gene expression analysis does not indicate significant increases in NOS2A or NOS3 gene expression, cellular NOS activities increased in Rotated 3-D cultures presumably through post-translational events. Furthermore, this increase in iNO and iROS occurs in all 3 CRC tested to date and within 10 min of entering Rotated 3-D culture.

In addition, we have now measured the activity of the NOS's that are expressed in MIP-101 cells in Rotated 3-D cultures compared to their activities in Static 3-D culture. Our gene expression analysis reveals that brain NOS is missing in these human colon carcinoma cells but that both inducible NOS (iNOS) and endothelial (eNOS) are expressed but that the gene and total protein expression of each NOS is not increased with change in cell shape and exposure to fluid shear stress (data not shown). However, when the activity of NOS's was analyzed, it is clear that NOS activity significantly increases in Rotated 3-D culture. This is critical information because it suggests that the activity of both NOS's are modified by post-translational modification which is somewhat novel for iNOS but not for eNOS. However, these data support the concept of targeting NOS function with inhibitors to decrease cell death that is described below.

5. Inhibition of NOS but Not General Antioxidant Therapy Inhibits Apoptosis: Two strategies have been tried to inhibit the apoptosis observed in the rotated RWV cultures. First, we tried N-acetylcysteine (NAC) pretreatment with or without NAC during the incubation in the RWV. NAC did not prevent apoptosis -even with pretreatment and appeared to have a weak but significant effect that increased apoptosis. However, 5mM NMMA a general NOS inhibitor significantly inhibited apoptosis as measured by Annexin V staining. While the amount of staining is only partially decreased from ~80% of the untreated 3-D Rotated culture to about 40% in the NMMA treated rotated culture, it is a significant effect. Interestingly, there appears to be another mechanism causing apoptosis that is involved because the NOS inhibition blocks only about half of the apoptosis caused by rotation compared to the low 10-15% apoptosis observed in either 2-D Monolayer or 3-D Static cultures.

6. Growth in 3-D for 24 Hours Alters Gene Expression in MIP-101 Cells. In collaboration with Dr. L. Augenlicht and his group at Montefiore Hospital in New York, we have examined the gene expression profiles of weakly metastatic MIP-101 cells cultured for 1 to 4 days under Monolayer 2-D, Stationary 3-D and Rotated 3-D conditions and then compared the gene expression of MIP-101 cells in the two 3-D cultures to the Monolayer 2-D pattern in custom arrays that contained > 17,000 genes. The expression of 253 genes was increased or decreased 4-fold or greater when 24 hr cultures of Rotated 3-D cells were compared to Static 3-D (Data Not Shown). We also present a number of important genes involved in apoptotic and oxidative stress pathways. For instance, the expression of IL-6 and Bcl-2 is decreased while the expression of gp130 and IL-6Ra are increased and the expression of NOS2A and NOS3 are not significantly changed (Data not shown). This last is important because it indicates that iNO increases as a result of the increase in activity in NOS's rather than because the expression of NOS isoforms is increased.

7. Inhibition of specific NOS Isoforms: The importance of eNOS: We initiated siRNA approaches to silence NOS2A and NOS3 expression in MIP-101. In order to confirm the importance of human NOS isoforms, we 1) identified that NOS1 was not expressed but NOS2A and NOS3 were by gene expression array (Data not shown), 2) used siRNA to NOS2A and NOS3 designed through the Ambion website as well as a Smart Pool NOS2A siRNA from Dharmacon to silence NOS mRNA. Transfection was optimized first with Lipofectamine 2000 using the siGLO Lamin and siGLO RISC-free siRNA from Dharmacon. Lipofectamine 2000 achieved an 85% transfection rate in semi-confluent cultures in 10 cm Petri dishes with 33 µl of lipid and 100 nM of siRNA that was ~15% more than Fugene 6. We then transfected MIP-101 cells with a negative control siRNA (Ambion), the NOS2A and NOS3 siRNAs (Ambion) and the Smart Pool siRNA (Dharmacon) along with untreated control cells and a Lipofectamine 2000 alone control. After transfection MIP-101 cells were cultured for 24 hr in monolayer cultures, harvested by trypsinization and placed on PolyHEMA-coated 10 cm culture dishes to increase iNO and iROS by static 3-D growth. Cells were collected 24 hr later (48 Hr after transfection) and the expression of targeted protein analyzed by western blot while iNO and iROS levels were determined by static fluorometry. The results indicate that 1) siRNA treatment decreased the specific targeted NOS protein level by at least 50% at 48 hr (although we need to use lower concentrations of siRNA to reduce possible off target effects) and 2) treatment with NOS3 but not NOS2A siRNA decreases iNO. Thus, eNOS (NOS3) may be important for iNO production during shape change and exposure to laminar flow shear stress, possibly because eNOS may be associated with mitochondria (60).

We then tested the postulates that 1) iNO was regulated by eNOS and not iNOS in Rotated 3-D cultures and 2) that reduction of iNO may increase viability of MIP-101 cells in rotated 3-D cultures. Interestingly, NOS3 siRNA treatment significantly inhibits iNO, iROS and mitochondrial membrane potential compared to treatment with NOS2A siRNA, and the other controls. Although inhibition of iNO and iROS by NOS3 siRNA was associated with a significant decrease in viability in Rotated 3-D culture, this may not be a specific effect of eNOS inhibition since Lipofectamine alone also reduced viability – even though treatment with NOS2A siRNA did not. Thus, further experiments are necessary to distinguish nonspecific or off-target effects of siRNA treatment must be done. However, our data indicate that iNO is regulated by eNOS in human MIP-101 CRC in 3-D culture and that commensurate changes occur in iROS and Mitochondrial Membrane Potential.

8. Use of fluorometry to measure NO in PBMC: In order to begin to align our work with the critical path for NASA research, we began to study PBMC from normal donors in the RWV to determine whether they responded the same way to simulated microgravity as the colon carcinoma cells do. Normal PBMC were isolated from blood by Ficoll-Hypaque density gradient centrifugation and cultured in the RWV for 24 hr. As a control, aliquots of these PBMC were also cultured as Static 3-D since these cells do not adhere to the culture substrate. We also studied PBMC within whole blood cultures in order to evaluate a fluorometer that we propose for use on ISS. The findings were 1) intracellular NO increases in PBMC within 2 hours and 2) remains greater than in the stationary controls during the first 20 hours, 3) similar increases in diluted whole blood occur in rotated cultures that is also higher than stationary controls for 24 hours, 4) intracellular ROS (measured with CM-H2DCFDA) also increases in whole blood (data not shown). These results

confirm the use of fluorometry because acute exposure to modeled microgravity significantly increases intracellular NO in PBMC and whole blood. Furthermore, the response of mononuclear cells in the presence of erythrocytes within the whole blood sample suggests that PBMC in the peripheral blood may also undergo similar nitrosative stress in crew in low earth orbit and that erythrocytes do not scavenge the NO which suggests that it is produced intracellularly. Thus, whole blood is an appropriate sample to study for countermeasure development. Furthermore, since similar increases in NO and ROS occur with such deleterious effects in epithelial cells, it is likely that these increases in NO and ROS mediate the inhibition of lymphocyte function seen in microgravity.

This is a straightforward extension of our work of Laguinge et al. (MS 2). The general inhibitor of NOS L-NMMA (NG-monomethyl-L-arginine) is used to inhibit the 3 NOS isoforms and has been used in humans to assess short-term effects of NOS inhibition on vascular resistance. Thus, this compound can be obtained under IND from the FDA for use in humans. However, for the types of in vitro studies anticipated in this study the L-NMMA will be available from commercial sources without restriction. As we have previously shown (MS 2), 5 mM L-NMMA added concurrently with exposure of MIP-101 cells to the RWV significantly reduces NO and iNOS levels, ß-tubulin degradation and apoptosis. We have also found that similar concentrations of 5 –10 mM L-NMMA inhibit intracellular NO and ROS in PBMC. Since we have shown that NO is increased within PBMC subjected to 20 hr in the RWV compared to static cultures of PBMC and that L-NMMA inhibits NO levels almost to the levels within static PBMC, we expect that this countermeasure will prevent apoptosis that is associated with elevation of cellular NO within PBMC. Direct detection of apoptosis in whole blood has been difficult because erythrocyes block detection of Annxin V-stained cells in whole blood imaged by flow cytometry (data not shown).

These results suggest that inhibition of NOS may provide part of an effective countermeasure for the early phase of exposure to hyporgravity. Interestingly, preliminary experiments with 100 nM 1400W, a specific inhibitor of iNOS, achieved the same improvement in cell survival as NMMA did. Taken together, these results support our postulate that iNOS and protein nitration are a cause of apoptosis observed in the Rotated 3-D cultures in the RWV.

9. CEA Expression in MIP-101 or Other CRC Inhibits Rotational Anoikis. Carcinoembryonic antigen (CEA) is a 180 kDa glycoprotein that is secreted by epithelial carcinomas – especially colorectal carcinoma. Previous work has shown that stable transfectants of MIP-101 that express full length CEA are metastatic when injected intrasplenically into athymic nude mice. We have also shown in an earlier NASA grant that CEA protein expression is increased in when MIP-101 cells change cell shape and then enter fluid shear stress since the MIP-101 cells increase CEA expression during growth in static 3-D as well as rotated 3-D cultures. We have even shown that CEA expression is increased during actual spaceflight (Jessup et al. Microgravity culture reduces apoptosis and increases the differentiation of a human colorectal carcinoma cell line. In Vitro Cell Dev Biol-Animal 36:367-373, 2000). Therefore, we have now begun to assess whether CEA expression is associated with reducing cell death in response to simulated microgravity. The stable CEA transfectant Clone 6 MIP-101 is highly metastatic because 50% of nude mice developed liver colonies whereas the other stable transfectant Clone 8 MIP-101 is also metastatic because it forms liver colonies in 25% of nude mice. Clone 8 forms more liver colonies than the parental MIP-101 which produces liver colonies in only 5% of recipients. Since CEA expression is associated with increasing metastatic potential in the clones and appears to be associated with survival in rotated 3-D cultures, we first examined the expression of CEA and CEACAMs (CEACAMs are the 28 other members of the CEA family – some of which are also expressed in PBMC) in the various CRC lines. As shown in the western blot of protein extracts probed with a monoclonal antibody that reacts with multiple CEA family members. MIP-101 expresses CEA when it is grown under 3-D conditions without rotation (Static 3-D) for either 24 or 48 hr (1 or 2 D) but not in 2-D monolayer culture (ML); Clone A, another poorly differentiate weakly metastatic CRC, expresses only CEACAM6 – 55 kDA, whereas CX-1 expresses CEA (185 kDa), CEACAM1 (160 kDa) and CEACAM6 (95 and 55 kDa). Interestingly, stable transfection of full length cDNA to CEA enables both Clone 6 and 8 to express CEA, CEACAM1 and CEACAM6. In addition, all 5 CRC lines express CEAR with dominant intranuclear expression in all lines and variable expression in the cytoplasm and plasma membrane. Clone 8 MIP-101 and CX-1 both localize CEA and CEAR at intercellular membranes whereas CEA is not associated with intercellular contacts in Clone 6 MIP-101. This appears to be the only difference in expression of CEA between Clone 6 and Clone 8 of MIP-101 and suggests that the interaction between CEA and CEAR in plasma membranes is important to the survival advantage that both Clone 8 MIP-101 and CX-1 have in rotated 3-D cultures.

L-NMMA increased the viability (Annexin V-/PI- subset) of Clone 6 in Rotated 3-D cultures by 26% but only increased the viability of Clone 8 cells by 7 % - even though the Clone 8 cells reduce intracellular NO concentration with L-NMMA treatment. This difference is likely secondary to the fact that more Clone 6 cells die in rotational 3-D culture than Clone 8 cells so that the effect of NOS inhibition is more apparent.

10. Transient Expression of CEA Inhibits Rotational Anoikis: Since the overexpression of CEACAM6 and CEACAM1 may also affect anoikis and the stable transfectants overexpress these members of the CEA family, we assessed whether transient transfection of full length CEA cDNA into MIP-101 may inhibit rotational anoikis. In this experiment, MIP-101 cells were transfected with full length CEA cDNA in a pBluescript KS(+) plasmid or with empty vector. At 24 hr cells were harvested and then put into the Rotational culture for another 24 hr whereupon they were harvested for analysis of apoptosis by flow cytometry as well as CEA expression by Western blot. The results demonstrate that 1) Transient transfection induces expression of only CEA and not CEACAM1 or CEACAM6 (data not shown) and 2) transient transfection and expression of CEA significantly reduces rotational cell death. In addition, we have also found that CEA expression is induced in MIP-101 cells cultured in Static 3-D culture for 24 hr and that this “preconditioning” also increases survival in Rotated 3-D culture compared to MIP-101 cells that were cultured in Monolayer 2-D culture that does not increase CEA expression (data not shown). The effect of transient transfection with CEA is associated with the expression of only CEA and not CEACAM1 or CEACAM6 (data not shown).

5. Future Plans: Dr. Augenlicht's and my original concerns were to examine gene expression at 4 - 6 days of rotational culture. However, most of the dynamic changes that occur within these cells that lead to cell death really occur during the first 24 hr. As a result, we are now examining gene expression changes during the first 24 hr to determine whether there will be specific alterations in the expression that are associated with apoptosis. Also since we have now identified that CEA protein expression reduces cell death in rotational culture, we are examining the gene expression of MIP-101 stable CEA transfectants during the first 24 hr of 3-D culture to determine whether CEA affects specific gene expression. We will do this during the next 5 months of no cost extension. Dr. Augenlicht joins me in expressing enthusiasm for the alteration in gene expression that is seen in these cultures and we hope in the next 5 months to have a coherent analysis of the gene changes during acute exposure to modeled microgravity.

Bibliography Type:	Description: (Last Updated: 11/27/2020)
Abstracts for Journals and Proceedings	J.M. Jessup "What Should I Have Known About Doing Cell Biology Experiments in Space?" Presented at the Pre-Meeting Workshop of the American Society of Gravitational and Space Biology in Huntsville, AL Nov, 12, 2003 Proc ASGSB, 2003 , Nov-2003
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Awards	Life Sciences Directorate, NASA to J. M. Jessup "Special Space Flight Achievement Award from the Space and Life Sciences Directorate for the Expedition 7 CBOSS-FDI Team" May-2004
Awards	Life Science Directorate, NASA to J. M. Jessup "Special Space Flight Achievement Award from the Space and Life Sciences Directorate for the Expedition 8 CBOSS-FDI Team." Sep-2004
Presentation	J. M. Jessup "Astronaut Training for CBSS-FDI For CY 2003-2004 to M. Fincke" Training for CBOSS-FDI Feb-2004
Presentation	J.M. Jessup "Astronaut Training for CBSS-FDI For CY 2003-2004 to C.M. Foale" Crew debrief for CBOSS-FDI Jul-2004